

Rules for Naming Variables in R

1. Syntactic Rules (Language Requirements)

In R, a variable name (identifier) must follow strict syntax rules to be considered **syntactic** (valid without special escaping).

Core Syntax

- **Allowed Characters:** Letters (a-z, A-Z), digits (0-9), the period (.), and the underscore (_).
- **First Character Rule:** A valid variable name must start with either a **letter** or a **period** (.).
- **The Period Constraint:** If a variable name begins with a period (.), the second character **cannot be a digit**.
 - Variables starting with a period are treated as **hidden objects** in the R workspace (they will not appear in `ls()` unless `all.names = TRUE` is set).
- **Forbidden Leading Characters:** A variable **cannot** start with a number or an underscore (_).
- **Case Sensitivity:** R is strictly case-sensitive. `my_data`, `My_Data`, and `MY_DATA` refer to three entirely separate variables.

2. Examples of Valid vs. Invalid Names

Variable Name	Status	Reason / Explanation
<code>user_age</code>	Valid	Uses letters and an underscore.
<code>.dataset2</code>	Valid (Hidden)	Starts with ., followed by a letter.
<code>sales.2026</code>	Valid	Uses letters, a period, and numbers.

Variable Name	Status	Reason / Explanation
2026_sales	Invalid	Starts with a digit.
.2026_sales	Invalid	Starts with ., but the second character is a digit.
_user_id	Invalid	Starts with an underscore (_).
user-name	Invalid	Hyphens (-) are parsed as subtraction operators.
total_\$ \vert{} Invalid \vert{} \$ is a reserved symbol for list/data frame indexing.		

3. Reserved Keywords

You cannot use R's built-in reserved words as standard variable names.

```
if      else      repeat   while      function  for
in      next      break     TRUE       FALSE     NULL
inf     NaN       NA        NA_integer_ NA_real_
NA_complex_ NA_character_
```

Attempting to assign values directly to constants like `TRUE = 5` or `NULL = 10` will trigger an immediate syntax error.

4. Non-Syntactic Variable Names (Backticks)

If you must use special characters, spaces, or start a variable with a number (common when reading external Excel files or databases), you can enclose the name in **backticks** (```):

```
R
# Non-syntactic variables defined via backticks
`2026 Total Revenue` <- 150000
`user-name` <- "Alex"

# Accessing non-syntactic variables
print(`2026 Total Revenue`)
```

Note: While supported, non-syntactic names are discouraged in standard workflows because they require continuous backtick escaping.

5. Community Naming Conventions & Style Guides

While valid syntactically, choosing a consistent naming style improves code readability and maintainability.

Recommended Style: `snake_case`

The **Tidyverse Style Guide** (widely adopted across the R community) strongly recommends using lower-case letters, numbers, and underscores:

- **Recommended:** `patient_id`, `calculate_mean()`, `model_fit_2026`
- **Avoid:** `PatientID` (PascalCase), `patientID` (camelCase)

Why Avoid `period.style` Variables?

Older R code frequently used periods to separate words (e.g., `data.frame`, `as.numeric`). In modern R, this practice is discouraged for variable naming because the period is the default delimiter for **S3 Object-Oriented method dispatch** (e.g., `print.default()`, `summary.lm()`). Using periods in object names can create ambiguity between function methods and object names.

Quick Reference Summary

1. **Start with:** A letter (or a period followed by a letter).
2. **Contain only:** Letters, numbers, `_`, and `..`
3. **Avoid:** Starting with numbers/underscores, reserved words, spaces, and hyphens.
4. **Best Practice:** Use `snake_case` for all variable and column names. Normalize this format.